PROJECT ON

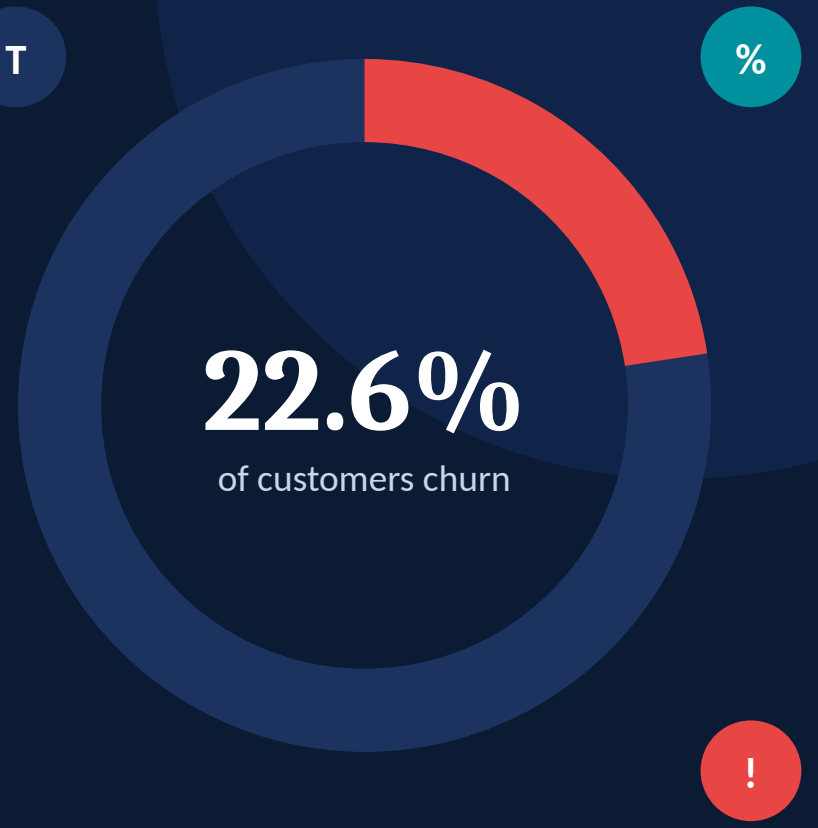
Telecom Customer Churn Analysis

From Dirty Data to Business Insight - A Full EDA Journey

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5,500	12	10+	22.6%
Raw Records	Columns	Issues Cleaned	Churn Rate

PROJECT BRIEF

Problem Statement & Objective

Problem Statement

Telecom providers lose significant recurring revenue when customers churn - but raw customer data is rarely clean enough to model directly.

Nulls, outliers, inconsistent formatting, and duplicate records obscure the real signals driving customer attrition.

Objective

Clean a genuinely messy telecom customer dataset end-to-end and perform exploratory analysis to uncover churn drivers.

Quantify high-risk customer segments and translate findings into revenue-at-risk figures that support retention decisions.

DATA UNDERSTANDING

Dataset Overview

Telecom Customer Churn Dataset

5,500
raw rows collected

12
features + 1 target

5,463
rows after cleaning

Numerical Columns

TenureMonths
Months as an active customer

MonthlyCharges
Current monthly billing amount

TotalCharges
Cumulative amount billed to date

SeniorCitizen
Binary flag (0/1)

Categorical Columns

Gender, ContractType
Customer demographic & plan type

InternetService
DSL / Fiber Optic / No Internet

TechSupport, PaperlessBilling
Service subscription flags

PaymentMethod, Churn
Billing method & target label

Target Column: Churn

Binary label (Yes / No) - indicates whether the customer discontinued service. Used to drive every downstream cleaning, EDA, and modeling decision.

DATA CLEANING

Duplicate Records Found & Removed

A duplicate is an exact copy of another row: same CustomerID and same value in every column. Sample pairs from the raw data:

37
Exact duplicate rows

74
Rows involved (37 pairs)

5,500 → 5,463
Rows before and after

0
Duplicate CustomerIDs after

Row	CustomerID	Gender	TenureMonths	ContractType	MonthlyCharges	PaymentMethod	TotalCharges	Churn
307 (kept)	1626-MZEYG	NaN	9	Month-to-Month	NaN	Electronic Check	707.98	No
5383 (dropped)	1626-MZEYG	NaN	9	Month-to-Month	NaN	Electronic Check	707.98	No
465 (kept)	1638-MXRBY	Male	13	One Year	45.16	NaN	591.86	No
1512 (dropped)	1638-MXRBY	Male	13	One Year	45.16	NaN	591.86	No
3935 (kept)	9412-BVHGH	Female	19	One Year	NaN	Electronic Check	1103.8	No
5053 (dropped)	9412-BVHGH	Female	19	One Year	NaN	Electronic Check	1103.8	No
3251 (kept)	9903-FZAJV	Female	42	Month-to-Month	$18.00	Mailed Check	741.05	No
3497 (dropped)	9903-FZAJV	Female	42	Month-to-Month	$18.00	Mailed Check	741.05	No

Showing 4 of the 37 duplicate pairs and 8 of 12 columns. Each pair keeps the first row and drops the second using drop_duplicates().

DATA CLEANING

Column-by-Column Cleaning Summary

Column	Issue Identified	Cleaning Action
Full Dataset	37 exact duplicate rows	Dropped via drop_duplicates()
Gender	Casing dupes + 109 nulls	Standardized casing, imputed with mode
TenureMonths	Outliers: -3, 150, 999	Capped to valid range 0-72 (domain logic)
ContractType	Whitespace + null-like strings + 65 nulls	Stripped, converted, imputed with mode
InternetService	Null-like strings + 66 nulls	Converted to NaN, imputed with mode
TechSupport / PaperlessBilling	137 / 81 nulls	Imputed with mode
PaymentMethod	Casing dupes + 228 nulls	Standardized casing, imputed with mode
MonthlyCharges	Currency text + outliers + 109 nulls	Stripped "$", IQR-capped, median imputed
TotalCharges	Blank strings for tenure=0	Converted to numeric, structural 0-impute

DATA CLEANING

Raw vs. Cleaned Data - Side by Side

Same 5 customer records, shown before and after the cleaning pipeline. Highlighted cells mark the exact issues fixed.

BEFORE

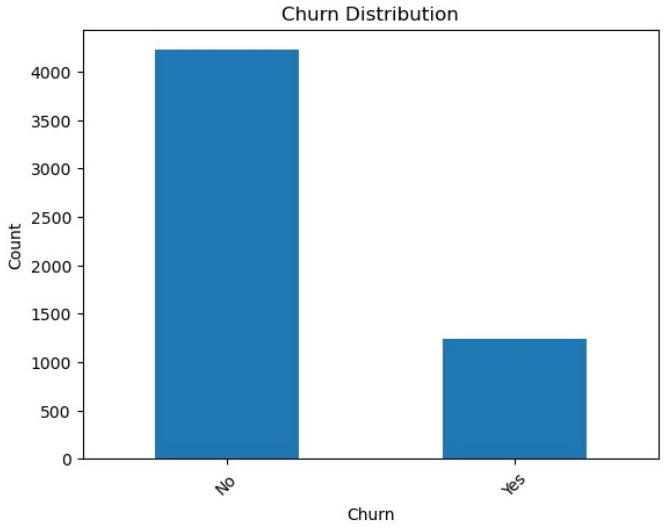
CustomerID	Gender	TenureMonths	ContractType	MonthlyCharges	PaymentMethod	Churn
3125-RGYHM	Male	37	One Year	$18.00	Mailed Check	No
7458-IKKLB	Female	39	One Year	$58.25	Electronic Check	No
3450-VRXAF	nan	24	Two Year	62.18	Mailed Check	No
8510-VCFZV	nan	51	One Year	97.78	Credit Card	No
1623-VTNOC	Female	150	Month-to-Month	87.25	Bank Transfer	Yes

AFTER

CustomerID	Gender	TenureMonths	ContractType	MonthlyCharges	PaymentMethod	Churn
3125-RGYHM	Male	37	One Year	18.0	Mailed Check	No
7458-IKKLB	Female	39	One Year	58.25	Electronic Check	No
3450-VRXAF	Female	24	Two Year	62.18	Mailed Check	No
8510-VCFZV	Female	51	One Year	97.78	Credit Card	No
1623-VTNOC	Female	72	Month-to-Month	87.25	Bank Transfer	Yes

EXPLORATORY DATA ANALYSIS

Univariate Analysis: Target & Numeric Distributions

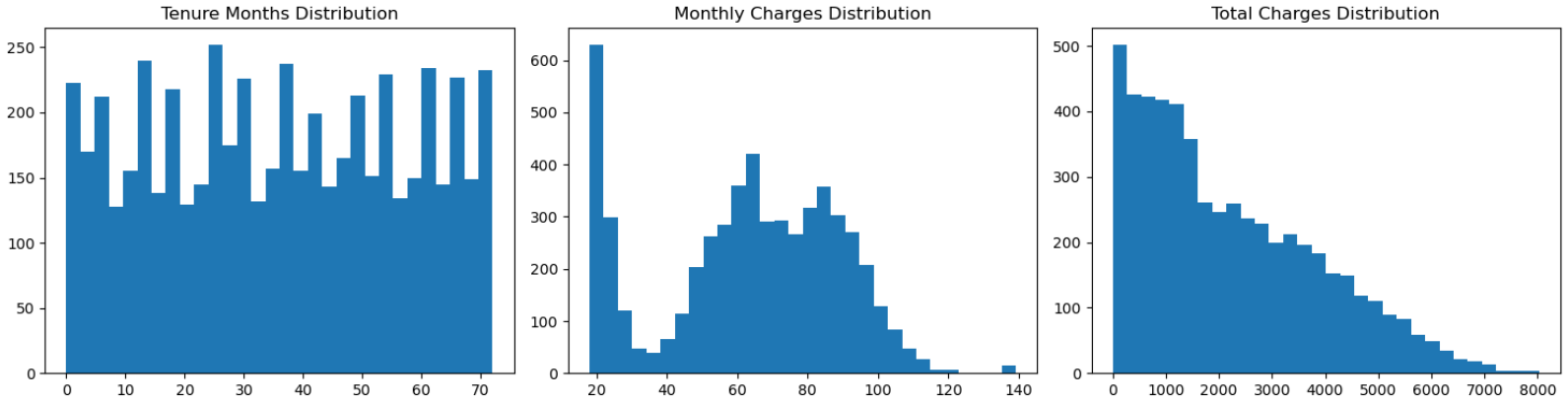


What this tells us

77.4%
Retained Customers

22.6%
Churned Customers

A healthy real-world class imbalance, common in churn problems. It shaped the modeling approach: evaluate with precision-recall rather than accuracy alone.



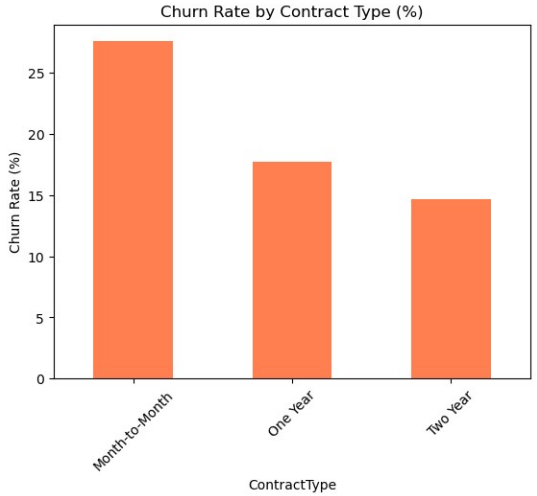
Observation:

Tenure spans 0-72 months, Monthly Charges center near $45-85, and Total Charges is right-skewed, as expected after capping.

BIVARIATE ANALYSIS

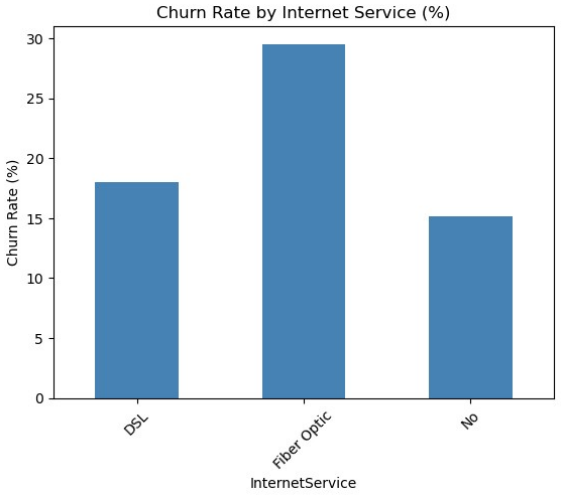
Churn Drivers: Contract Type & Internet Service

Churn Rate by Contract Type



Month-to-Month	**~27%**
One Year	~18%
Two Year	~15%

Churn Rate by Internet Service



Fiber Optic	**~29%**
DSL	~18%
No Internet	~15%

Insight

Month-to-Month customers churn at nearly double the rate of Two Year customers, the strongest single churn driver. Short commitments mean low-friction exits.

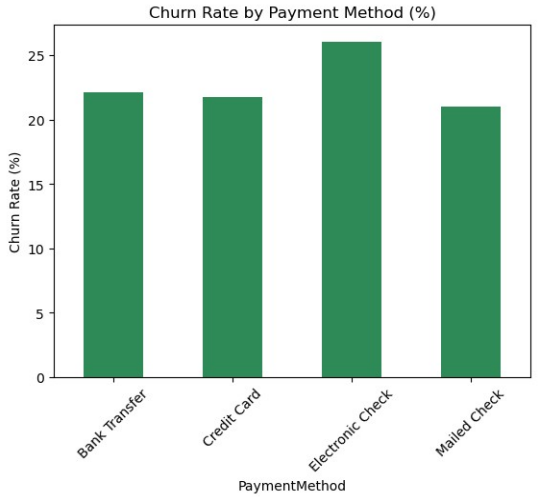
Insight

Fiber Optic customers churn far more than DSL or No Internet, likely from price sensitivity on the costliest tier. A priority segment for retention review.

BIVARIATE ANALYSIS

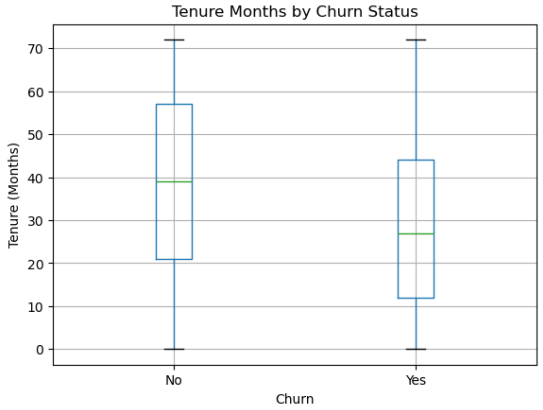
Churn Drivers: Payment Method & Numeric Features

Churn Rate by Payment Method

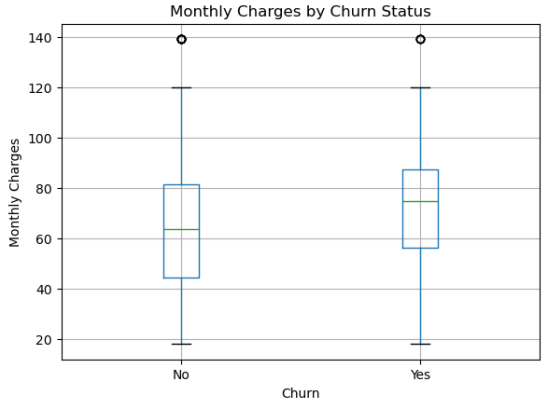


Electronic Check	**~26%**
Bank Transfer	~22%
Credit Card	~21.5%
Mailed Check	~21%

Numeric Features vs. Churn



Tenure vs. Churn



Monthly Charges vs. Churn

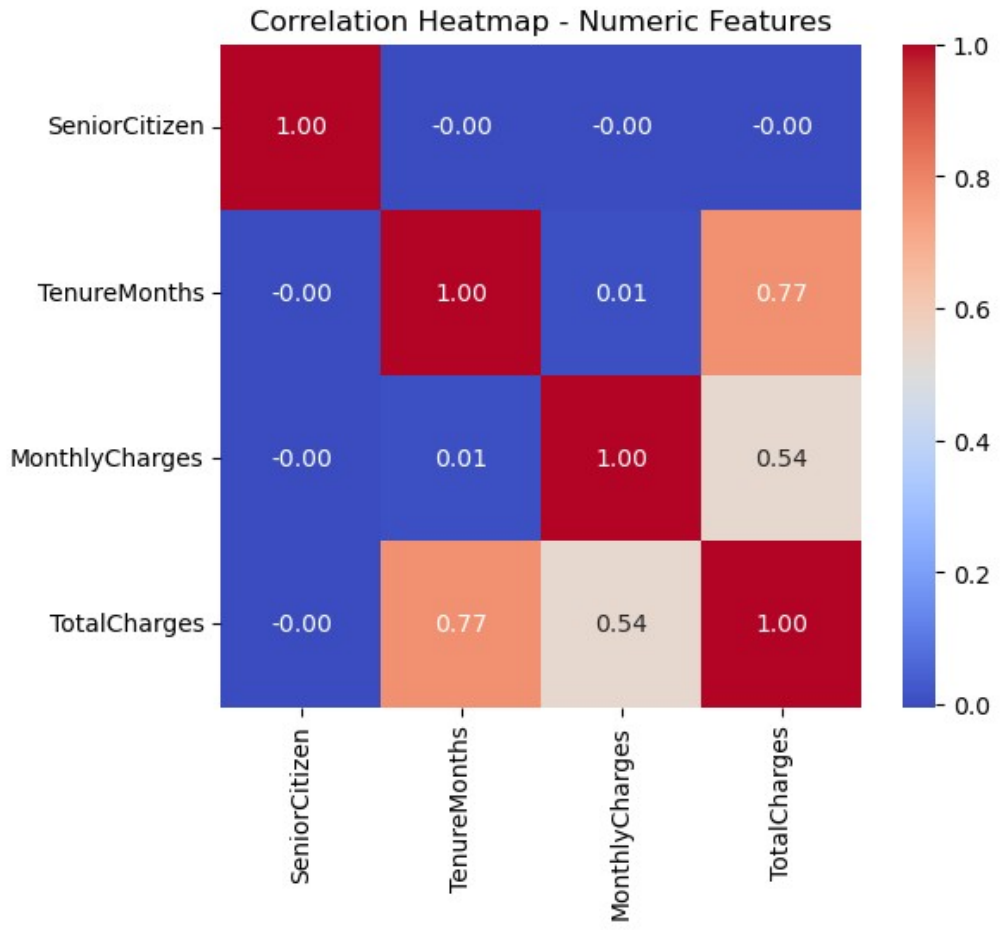
Insight

Electronic Check users churn more than automatic payment methods. Manual payers are less locked in, a well-known telecom churn pattern.

Observations

Tenure: churned customers skew to shorter tenure, so risk peaks early. Monthly Charges: churners carry slightly higher median charges, hinting at price sensitivity.

Correlation Heatmap - Numeric Features



Reading the Heatmap

TenureMonths and TotalCharges: strong positive correlation - expected, since longer-tenured customers naturally accumulate higher total billing.

MonthlyCharges and TotalCharges: moderate positive correlation, driven by the same underlying billing relationship.

SeniorCitizen: shows weak correlation with the other numeric features - behaves largely independently.

KEY FINDING

The High-Risk Customer Segment

What happens when the three strongest churn drivers combine?

Month-to-Month Contract + **Fiber Optic Internet** + **Electronic Check Payment** =

39.1%

churn rate in the high-risk segment

vs. 21.6% for everyone else - nearly 2x baseline

Why this matters

- Only ~6% of customers (335 of 5,463) fall into this segment
- Yet they account for a disproportionate share of total churn risk
- A precise, actionable target for retention campaigns - not a guess

BUSINESS IMPACT

Revenue at Risk

Turning churn probability into a dollar figure a business stakeholder can act on.

$86,284

Monthly revenue already lost to churned customers

$1.04M annualized

$27,974

Monthly revenue exposure from the high-risk segment alone

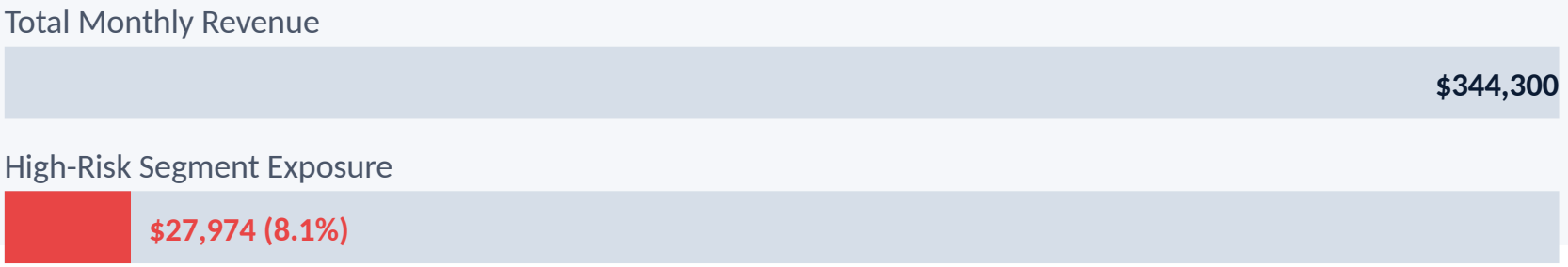
$335,692 annualized

$10,836

Already realized monthly loss from the high-risk segment

39% of segment revenue

Total Portfolio vs. High-Risk Exposure



8.1% of total revenue sits in a segment that's only 6% of customers

SUMMARY

Key Insights from EDA

1. **Overall churn rate stands at 22.6%**

   A healthy, realistic imbalance for classification modeling

2. **Month-to-Month contracts churn ~2x more than Two Year**

   Contract length is the single strongest churn predictor

3. **Fiber Optic customers show the highest churn among internet types**

   Likely price-sensitivity driven - a retention priority segment

4. **Electronic Check users churn more than automatic payment methods**

   Manual billing correlates with lower customer stickiness

5. **The combined high-risk segment churns at 39.1%**

   Nearly double the baseline - despite being only ~6% of customers

6. **8.1% of total revenue sits in the high-risk segment**

   A concentrated, actionable target for retention spend

Conclusion

What this analysis found

This EDA turned a messy telecom dataset into a clean, insight-rich foundation and revealed a precise high-risk churn segment (Month-to-Month + Fiber Optic + Electronic Check) with 39.1% churn and $27,974 in monthly revenue exposure.

Data Cleaning • Exploratory Data Analysis • Business Insight Generation

Thank You

Questions & Discussion Welcome

22.6%	39.1%	$27,974
Overall churn rate	High-risk segment churn	Monthly revenue at risk

SP

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Key Insights